Score Shaping and Pareto Optimality in Analog Integrated Circuit Design Automation

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***Abstract*—Despite increasing efforts to automate analog integrated circuit sizing, manual methodologies remain dominant due to their superior handling of design trade-offs. This work investigates the impact of score shaping functions on convergence speed and Pareto optimality in automated sizing. A case study on sizing a telescopic Operational Transconductance Amplifier (OTA) in the open-source IHP-130nm technology node using Differential Evolution demonstrates that the proposed sigmoid-based score normalization achieves faster convergence and improved design quality compared to traditional linear-based functions. Additionally, we present SpiceXplorer, an open-source, highly customizable circuit optimizer, and discuss its potential integration into future agentic workflows for fully automated design.**

***Index Terms*—Analog Integrated Circuits, Engineering Optimization, Sizing Automation, Evolutionary Algorithms, Operational Transconductance Amplifier**

I. Introduction

Analog integrated circuits (IC) are fundamental to modern electronics, yet defining the optimal specification for a given circuit block remains a context-dependent challenge [1]. For instance, analog frontends in biomedical applications prioritize critically low power consumption, high gain, and low input-referred noise over bandwidth [2]. Conversely, wireless receivers demand high bandwidth, moderate gain, and high linearity [3]. Consequently, the automation of analog circuit sizing is typically formulated as a constrained optimization problem, where the goal is to maximize a figure of merit (FOM) while adhering to strict performance boundaries.

Despite the surge of interest in automating this process using machine learning [4], large language models [5], and symbolic analysis [6], a critical gap remains in the literature. Research often over-fixates on improving the *solver*—the algorithm navigating the design space—while neglecting the *problem formulation*—the score function itself. This oversight is significant because the efficacy of any optimization algorithm is fundamentally bounded by the quality of the problem formulation. An ill-defined or arbitrary score function, such as one utilizing uncalibrated random weights or sharp, discontinuous penalty steps, can render even the most sophisticated black-box optimizer inefficient. If the mathematical "shape" of the objective function does not align with the solver's capabilities, the optimizer may stagnate in local optima or waste resources exploring infeasible regions. Currently, there is a lack of rigorous benchmarking on how different score shaping strategies interact with standard black-box solvers.

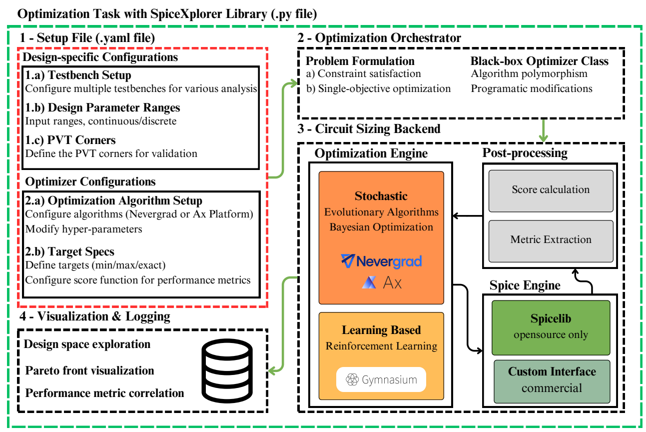


Fig. 1. The SpiceXplorer Python-based tool developed for this study [7]–[9]. It orchestrates the interaction between black-box optimizer and SPICE simulators to enable the systematic benchmarking of score shaping methods.

To address this gap, the contribution of our work is two-fold. First, we present *SpiceXplorer*, a highly customizable circuit optimization framework. As shown in Fig. 1, the tool utilizes a structured YAML configuration to orchestrate the interaction between black-box optimizers and SPICE simulators allowing for rapid prototyping of different problem formulations. The code is made available on GitHub[1] to encourage further innovation in the EDA community. Second, we utilize this framework to systematically study the effect of "Score Shaping." We provide an empirical analysis of how different normalization methods impact the speed of convergence and the quality of the Pareto front traced by the optimizer.

II. Related Work

While the majority of EDA literature contrasts optimization algorithms (e.g., "Algorithm A vs. Algorithm B"), a critical subset investigates the mathematical formulation of the objective functions that guide these solvers. Various strategies have been proposed to handle constraints, ranging from adaptive dynamic penalties [10] to product-based aggregation [11] and probabilistic feasibility scaling [12]. However, the most distinct shift in formulation comes from Moreto et al.

[1]https://github.com/REDACTED/REDACTED

[13], who argue that the *shape* of the penalty function is paramount. They propose replacing traditional step or linear functions with smooth Gaussian profiles. Unlike step functions, which offer binary feedback, or linear functions, which contain sharp discontinuities at the target boundaries, Gaussian functions provide a continuous gradient. This smoothness allows the optimizer to effectively rank solutions in critical transition regions—where a design is "almost" meeting specifications—rather than discarding them. Consequently, this approach was shown to significantly improve design yield and robustness to PVT variations, proving that the definition of "failure" is as important as the search algorithm itself. Moreover, the optimal choice depends heavily on external factors such as the design intent and the solver type (e.g., BFGS requires gradients, whereas Differential Evolution does not).

III. Method

SpiceXplorer treats circuit sizing as a search for a parameter vector $\mathbf{x}$ that satisfies a set of performance targets $\mathbf{T}$. To investigate the impact of score shaping, we formulate the problem as a CSP where the fitness function $F(\mathbf{x})$ is derived from aggregated normalized penalties. This approach—prioritizing feasibility before performance maximization—aligns with Rahimi *et al.* [14], who demonstrated that decomposing optimization into a feasibility phase followed by objective optimization significantly improves efficiency. Our score shaping strategies pursue this principle through continuous penalty design rather than explicit phase separation.

A. Raw Penalty Formulation

For each performance metric m_i, a raw penalty p_i is calculated based on its optimization type:

- **EXCEED**: $p_i = \max(0, T_i - m_i)$. E.g., DC gain, bandwidth.
- **MINIMIZE**: $p_i = \max(0, m_i - T_i)$. E.g., power, noise.
- **EXACT**: $p_i = |m_i - T_i|$. E.g., phase margin (60°).

B. Score Normalization Strategies

Raw penalties often span vast magnitudes (e.g., 10^9 Hz vs. 10^{-6} A). Effective normalization is required to prevent numerical bias. We compare two strategies:

1) Linear Normalization (Baseline): Penalties are normalized relative to a constant S_i, producing a unitless relative error $\hat{P}_i$:

$$\hat{P}_i = \frac{p_i}{S_i} \tag{1}$$

where S_i is the target magnitude $|T_i|$ or a user-specified tolerance. This allows the designer to weigh sensitivity; for example, equating a 500 MHz bandwidth deviation to a 100% power error. However, linear normalization allows $\hat{P}_i$ to grow unboundedly, potentially causing gradient explosion or dominating the aggregate cost function.

2) Sigmoid Shaping (Proposed): To mitigate outlier dominance, we propose a non-linear layer that squashes errors into a bounded range $[0, 1)$:

$$P_i = \frac{2}{1 + e^{-\alpha \hat{P}_i}} - 1 \tag{2}$$

where α determines the saturation rate. By bounding the penalty, extreme outliers (metrics far from compliance) cannot dominate the total fitness score. This ensures the optimizer maintains sensitivity to metrics closer to the feasibility boundary, facilitating balanced convergence.

C. Aggregation and Objective Handling

To enable optimization beyond constraint satisfaction, the framework utilizes a dual-mode single-objective formulation. The aggregate fitness $F(\mathbf{x})$ switches between penalty and reward modes:

$$F(\mathbf{x}) = \begin{cases} \sum w_i R_i(m_i) & \text{if } \sum P_i(m_i) \approx 0 \\ -\sum w_i P_i(m_i) & \text{otherwise} \end{cases} \tag{3}$$

where w_i are user-defined weights and R_i is the reward for exceeding specifications. This piecewise formulation forces the optimizer to prioritize the "feasible region" (driving penalties to zero) before attempting to maximize the Figure of Merit.

IV. Case Study: Telescopic OTA

In this section, we evaluate the efficacy of the proposed optimization score functions by applying them to the sizing of a telescopic OTA in the IHP 130nm BiCMOS technology. The schematic of the circuit, featuring an improved telescopic cascode topology with ideal biasing, is shown in Fig. 2. The performance targets, listed in Table I, were selected to be competitive with manual designs reported in [15]. To ensure the optimizer is not artificially constrained, the design space (Table II) was defined with broad parameter ranges. Future work could focus on strategies to better adjust the limits of these parameters based on domain knowledge [5].

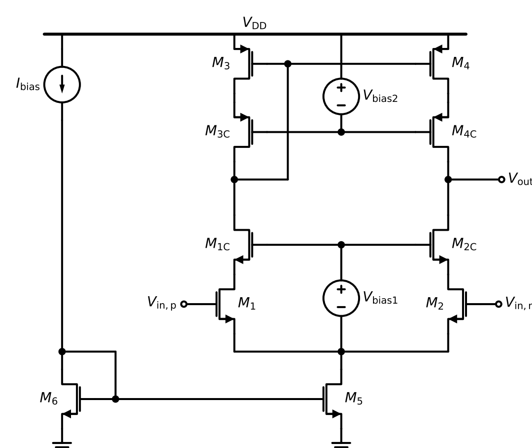


Fig. 2. Schematic of the improved telescopic cascode OTA with ideal bias sources used for the case study [15].

TABLE I
TARGET PERFORMANCE METRICS AND OPTIMIZATION OBJECTIVES
(NOISE INTEGRATED FROM 1 MHZ TO 10 MHZ)

Metric	Target	Unit	Reward Type
Unity Gain Frequency	>200	MHz	Maximize
DC Gain	>40	dB	Maximize
Integrated Input Noise	<1.2	mV	Minimize
Total Supply Current	<25	μA	Minimize
Phase Margin	$=60 \pm 10$	deg	Constraint Only
Settling Time	<15	μs	Constraint Only

A. Optimizer Setup

To rigorously benchmark the performance of the score functions, we employed three distinct search strategies:

- **Baseline (LHS):** A Latin Hypercube Sampling (LHS) method was used to generate a quasi-random baseline. This brute-force exploration was allocated a generous budget of 12,000 simulations to map the feasible regions of the design space comprehensively.
- **Gradient-Free Optimization (DE):** The Differential Evolution (DE) algorithm was selected as the primary global optimizer due to its robustness against the non-convex nature of analog design landscapes.
- **Gradient-Based Optimization (BFGS):** The Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm was included to evaluate the score functions' differentiability and performance with gradient-descent methods.

Both the DE and BFGS optimizations were restricted to a budget of 2,000 simulations to simulate a resource-constrained design environment. This constraint highlights the efficiency of the score shaping method in guiding the solver to a valid solution with limited evaluations.

TABLE II
DESIGN PARAMETER SEARCH SPACE FOR THE OTA WITH 14 DIMENSIONS (WIDTHS, LENGTHS, AND BIAS VOLTAGES).

Parameter	Symbol	Value / Range
Supply Voltage	V_{DD}	1.5V
Output Loading	C_{L}	50 fF
Reference Bias Current	I_{bias}	5 μA
Common-mode Input	$V_{\mathrm{in,p}}, V_{\mathrm{in,n}}$	800 mV
Cascode Bias	$V_{\mathrm{bias1}}, V_{\mathrm{bias2}}$	0 to V_{DD}
Transistor Widths	$W_{1-6}, W_{1\mathrm{C}-4\mathrm{C}}$	0.180 μm to 200 μm
Transistor Lengths	$L_{1-6}, L_{1\mathrm{C}-4\mathrm{C}}$	0.180 μm to 200 μm

B. Results & Discussion

Fig. 3 depicts the score trace for the DE optimization strategy using sigmoid and linear score functions. Although the absolute score magnitudes are not directly comparable due to normalization differences, the zero-crossing point universally indicates constraint satisfaction. The linear function never crosses zero. The sigmoid function not only meets the constraints but also accumulates rewards for optimizing specifications beyond the targets. In contrast, the LHS sampling method stagnates below -40 score points for the first 2000 iterations and only reaches -6 by the 10,000-simulation mark, failing to meet constraints. Similarly, the BFGS algorithm remains trapped at high penalty values for both score definitions. Consequently, these two non-converging engines are excluded from Fig. 3 to ensure visual clarity.

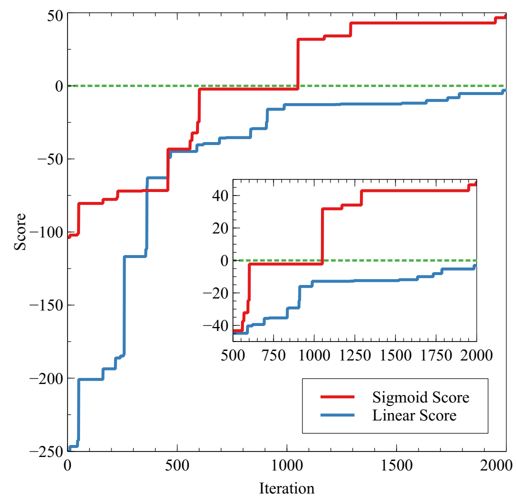
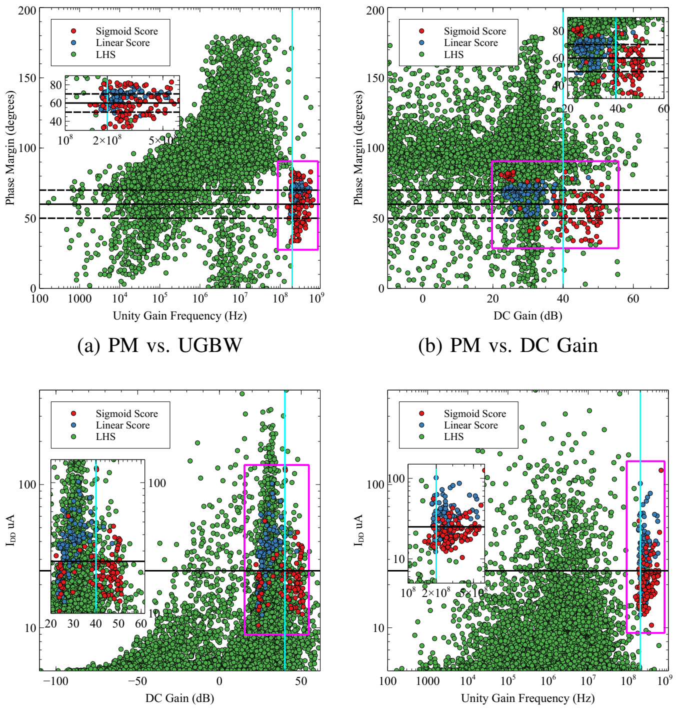


Fig. 3. The score function over 2000 optimization step. The sigmoid-based score function meets constraints after 500 iterations whereas the linear score never meets the constraints.

Fig. 4 superimposes the top 100 designs after optimization upon the baseline LHS search results. The performance space exploration suggests that while both optimizers gravitated toward the feasible region, the sigmoid-based optimizer demonstrated superior convergence. Specifically, Figure 4c and Figure 4d show that the optimal points identified by the sigmoid function correspond to lower power consumption with higher gain and bandwidth, indicating a more efficient design realization. We compare the best design identified by each method in Table III. The sigmoid-based optimizer consistently achieved better performance, with the exception of total integrated noise. Interestingly, the Linear score function failed solely on the total current consumption constraint ($43.7\mu A$ vs $< 25\mu A$). This violation explains the lower noise figure achieved by the linear method, as there is a fundamental trade-off between noise and power consumption. This outcome highlights the shortcoming of linear score definitions in multi-objective problems with varying scales; the large numerical magnitude of UGF (MHz) and Gain (dB) dominated the cost function, effectively masking the penalty violation for current constraints (μA).

The histograms in Fig. 5 depict the distribution of performance metrics for all designs visited during optimization. The y-axis represents the count normalized as a percentage of total simulations. The UGF histograms in Figure 5a demonstrate that the sigmoid function encouraged exploration of higher frequency ranges while maintaining diversity, implying the optimizer did not over-fit to a single high-UGF region.



(a) PM vs. UGBW (b) PM vs. DC Gain

(c) Current vs. DC Gain (d) Current vs. UGBW

Fig. 4. Optimization results for the Telescopic OTA, comparing the top 100 designs against the LHS baseline.

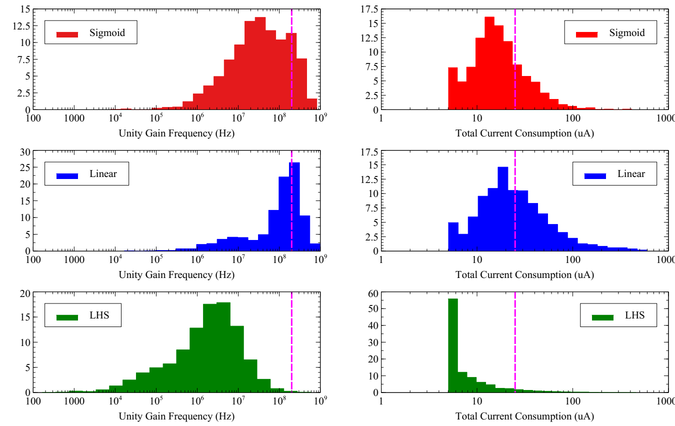
TABLE III
PERFORMANCE COMPARISON OF OPTIMIZATION SCORING FUNCTIONS

Metric	Unit	Sigmoid (Best)	Linear
Unity Gain Frequency (UGF)	MHz	**527.7**	236.5
DC Gain	dB	**49.3**	40.2
Phase Margin (PM)	°	**69.0**	70.0
Total Integrated Noise	mV	1.02	**0.56**
Total Current (I_{DD})	μA	**20.6**	43.7
Settling Time (t_{settle})	μs	**0.13**	0.29

Moreover, Figure 5b reveals how the optimizers navigated power consumption. The LHS distribution clusters heavily at low currents, suggesting designs trapped in sub-threshold operation near the $5\mu A$ reference bias. Conversely, both score definitions exhibited higher power consumptions implying they explored designs in linear or saturation mode of operation, with the sigmoid function tending more towards the low power consumption.

C. Impact of Score Shaping on Designer Intent

Our results highlight a critical divergence between mathematical optimization and engineering intent. The linear score definition frequently stalled because a single "outlier" metric (e.g., bandwidth far from target) dominated the aggregate cost function, masking the gradients of other sensitive metrics like power. By saturating these large errors, the Sigmoid function effectively creates a "soft" feasibility barrier. This formulation better captures the designer's intent: a solution that marginally meets *all* specifications is far preferable to one that maximizes a single metric while violating others. The Sigmoid shape



(a) Unity Gain Frequency (b) Total Current Consumption

Fig. 5. Histograms comparing simulation methodologies.

forces the optimizer to prioritize "balance" (feasibility) over "peaks" (maximization of a single variable), preventing the solver from wasting resources in invalid design regions.

V. FUTURE WORK: AGENTIC AI INTEGRATION

The next evolution of SpiceXplorer involves integrating Agentic AI workflows to act as a "silicon architect." While the current solver optimizes parameters, an agentic framework can supervise the entire design flow, including layout generation and parasitics extraction. Key enhancements include:

- **Knowledge-Driven Initialization:** Agents can leverage circuit theory to provide "warm-start" initializations, significantly accelerating convergence compared to random sampling [16].
- **Intelligent Constraint Synthesis:** Beyond simple sizing, agents can parse topologies to enforce symmetry and g_m/I_D consistency, or even propose topological variations (e.g., inserting Miller compensation) to meet stringent stability targets.
- **Resource-Aware Orchestration:** A "Manager Agent" can strategically coordinate multi-fidelity tools, evaluating candidate maturity before committing to expensive tasks like parasitic extraction [17].

VI. CONCLUSION

This work investigates the importance of score shaping in analog IC design automation. Through the sizing of a telescopic OTA in 130nm technology, we demonstrated that sigmoid-based normalization outperforms traditional linear penalties. The proposed method not only achieved constraint satisfaction where linear methods failed but also discovered designs with approximately 50% lower power consumption. These results highlight that robust problem formulation is critical for navigating design specifications. Future efforts will extend *SpiceXplorer* to support agentic workflows for topology-aware sizing.

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